MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FUSE, KEY CODE GZC

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic fuse module.

* 1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787	-	Modules, Standard Electronic, General Specification for.
MIL-F-23419/4	-	Fuses, Instrument Type, Style FM04 (Nonindicating).
MIL-F-23419/8	-	Fuses, Instrument Type, Style FM08 (Subminiature-High Performance).

STANDARDS

MILITARY

MIL-STD-1389	-	Design Requirements for Standard Electronic Modules.
MIL-STD-1665	-	Test Equipment for the Standard Electronic Modules Program.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

NAVORD Drawing 10001-2465390 - Contact, Electrical Connector.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.808 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be GZC.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A. The cross sectional area of the entire fin shall not exceed 57 percent of the area designated in MIL-STD-1389, figure 6.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/209-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.5.9 Fuse type. Fuses used in this module shall conform to the requirements of MIL-F-23419, style FM04 or FM08. For fuse lead configuration, see figure 1.

3.5.10 Fuse contacts. Fuse contacts used in this module shall conform to the requirements of NAVORD Drawing 10001-2465390, part 8.

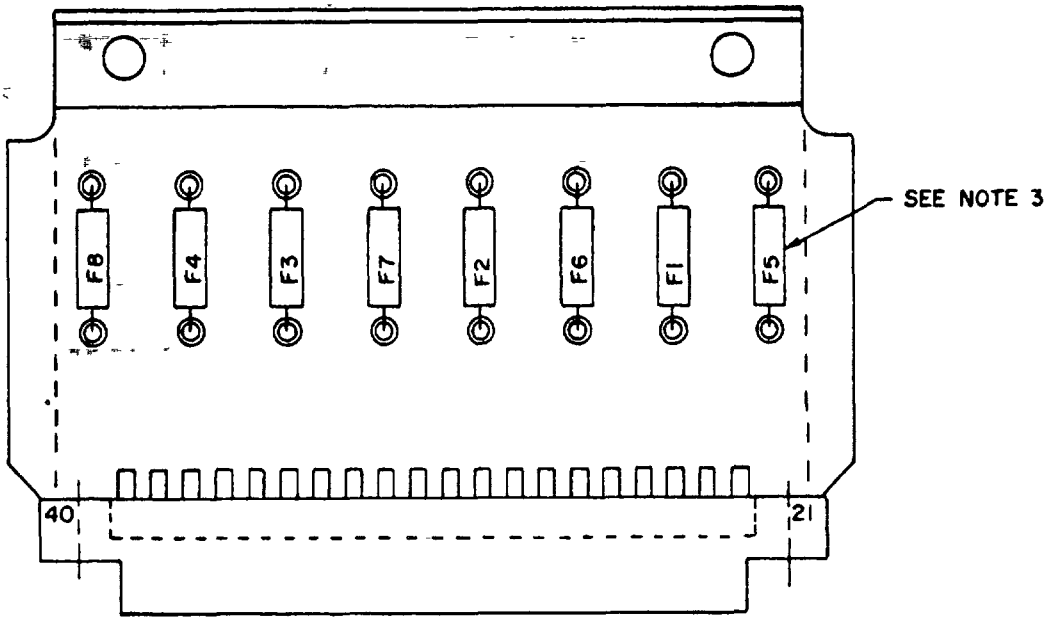
3.5.11 Location. The relative location of the fuses shall conform to figure 1. The fuse component identification and value shall be marked relative to each fuse position on the fuse mounting surface or printed circuit board.

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 2.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1, 2, and 3 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

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NOTES:
1. Fuse marking:
   F1 = 5 A.
   F2 = F3 = F5 = F8 = 3/4 A.
   F4 = 1 A.
   F6 = 1/4 A.
   F7 = 2 A.
2. Dimensions are in inches.
3. Lead configuration:

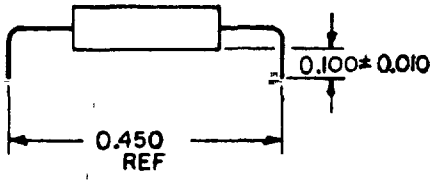


FIGURE 1. Module fuse configuration requirements..

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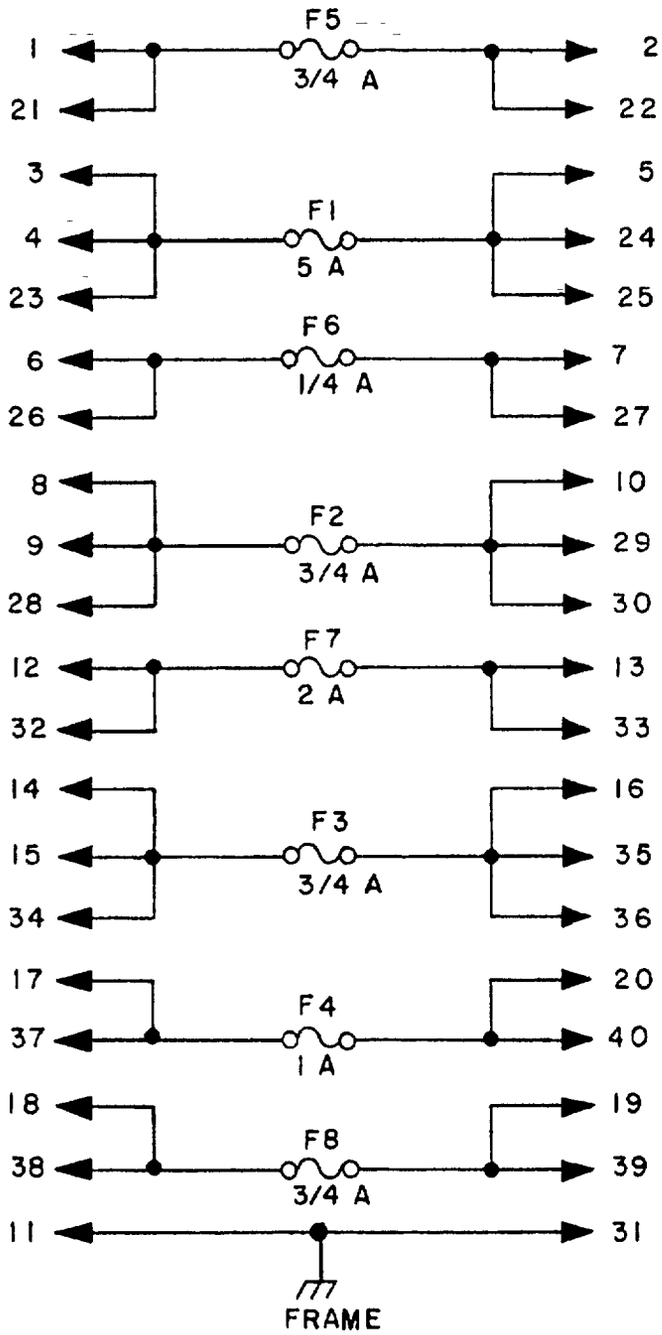


FIGURE 2. GZC module function diagram, fuse.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 2/-4/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A19	Isolation	II	X 1/	10	∞	1	∞	10	∞	1	∞	MΩ
B	Visual inspection	---	X	← See 4.6 →								
C1	Fuse isolation and integrity, F5	III	X	7.9	8.1	7.9	8.1	7.9	8.1	7.9	8.1	V
C2	Fuse isolation and integrity, F1	III	X	6.9	7.1	6.9	7.1	6.9	7.1	6.9	7.1	V
C3	Fuse isolation and integrity, F6	III	X	5.9	6.1	5.9	6.1	5.9	6.1	5.9	6.1	V
C4	Fuse isolation and integrity, F2	III	X	4.9	5.1	4.9	5.1	4.9	5.1	4.9	5.1	V
C5	Fuse isolation and integrity, F7	III	X	3.9	4.1	3.9	4.1	3.9	4.1	3.9	4.1	V
C6	Fuse isolation and integrity, F3	III	X	2.9	3.1	2.9	3.1	2.9	3.1	2.9	3.1	V
C7	Fuse isolation and integrity, F4	III	X	1.9	2.1	1.9	2.1	1.9	2.1	1.9	2.1	V
C8	Fuse isolation and integrity, F8	III	X	0.9	1.1	0.9	1.1	0.9	1.1	0.9	1.1	V
D1, D2, D5-D20	Continuity	IV	Q	0	1	0	1	0	1	0	1	Ω
D3, D4	Continuity	IV	Q	0.0	1.6	0.0	1.6	0.0	1.6	0.0	1.6	Ω
E1-E4, E6-E8	Voltage drop	V	Q	70	180	70	180	55	200	55	200	mV
E5	Voltage drop	V	Q	70	375	70	375	70	434	70	434	mV
F1, F2	Continuity	VI	S	0	1	0	1	0	1	0	1	Ω

1/ These parameters shall be 100 percent tested during module assembly before any fuses are installed (see 4.5.1 for qualification testing).
2/ "X" parameters shall be 100 percent tested during module acceptance tests.
3/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater at 25°C.
4/ "Q" parameters shall be tested during qualification testing only.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test condition A shall be performed during module assembly before any fuses are installed. Test conditions B through F shall be performed in any order (provided the requirements of test condition A have been met) with the following exceptions. If test condition B is part of a sequence of tests, the order of testing shall be as follows: B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Isolation test (test condition A). During qualification testing, one end of each fuse shall be disconnected while performing test condition A. Test condition A is required to be performed during the initial electrical tests as specified in 4.5.2.

4.5.2 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions B, C through F, and A (see 4.5.1) in table I shall be performed in this order using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through F shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.2.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.3 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, except test condition A, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.4 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests C through F shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through F shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

4.6 Visual inspection. Visually inspect the component side of each module to insure that each fuse location is fitted with a fuse of the properly marked current rating (see figure 1).

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8); however, replacement of fuses F1 through F8 and retesting is permissible.

* 6.3 Cross reference of classification. Part number M28787/209-1 supersedes part number M28787/209. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum voltage	125 V dc
Maximum power dissipation (all fuses carrying rated current)	1.36 W

* 6.5 Module function. The module contains eight fuses. The module is pin programmable to provide the following fusing: One 5 amperes, one 2 amperes, one 1 ampere, four 3/4 ampere, and one 1/4 ampere.

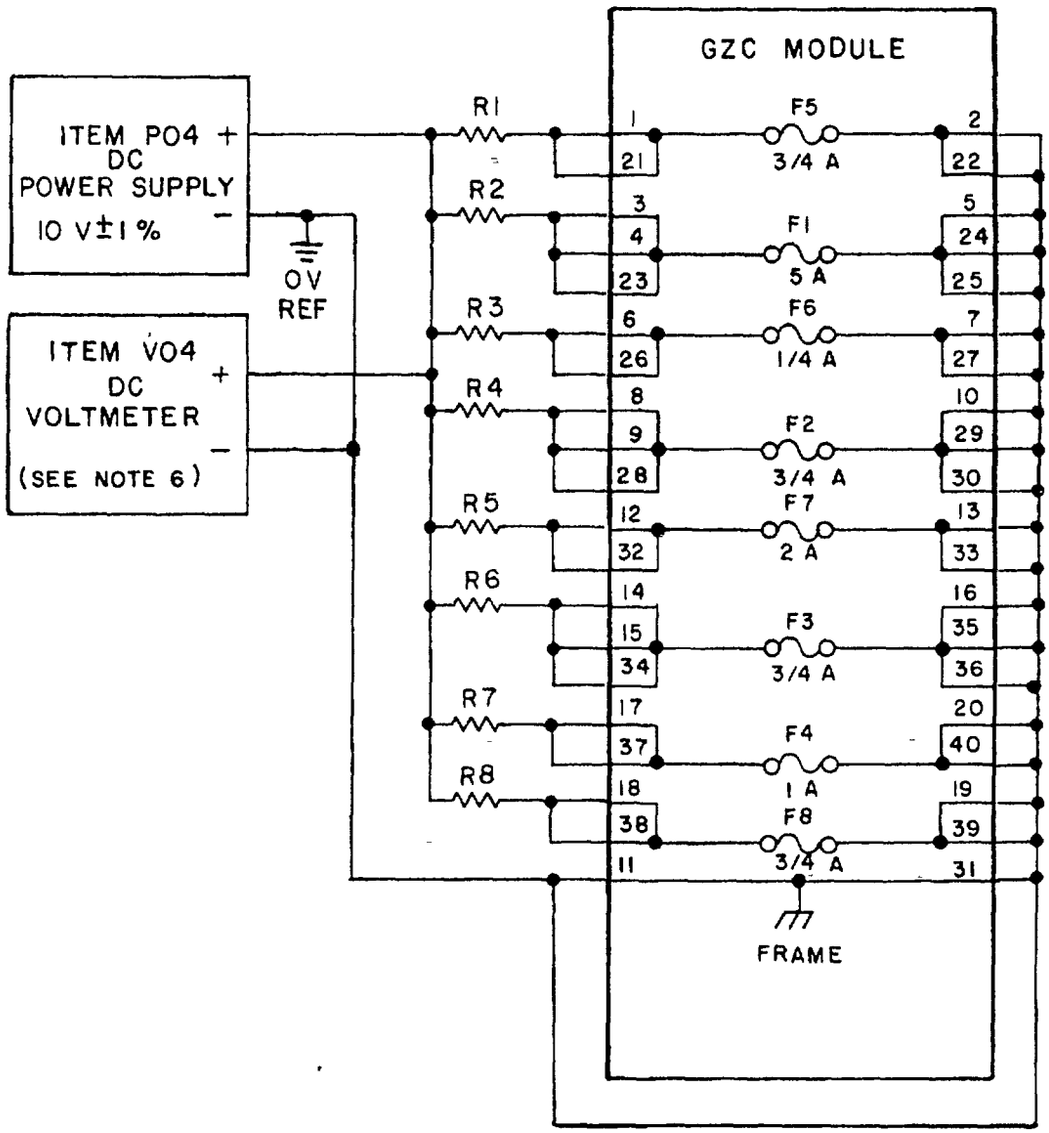
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective ot the marginal notations and relationship to the last previous issue.

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NOTES:
1 R1 = R4 = R6 = R8 = 17.8Ω ±1%, 10 W.
2. R2 = 2.6Ω ±1%, 50 W.
3. R3 = 53.3Ω ±1%, 5 W.
4. R5 = 6.7Ω ±1%, 20 W.
5 R7 = 13.4Ω ±1%, 10 W.
6 Item VO4 is used for equipment setup only.

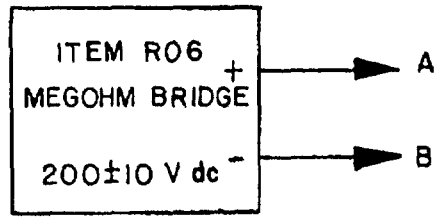
* FIGURE 3. Operating temperature, component temperature, vibration, and life test circuit.

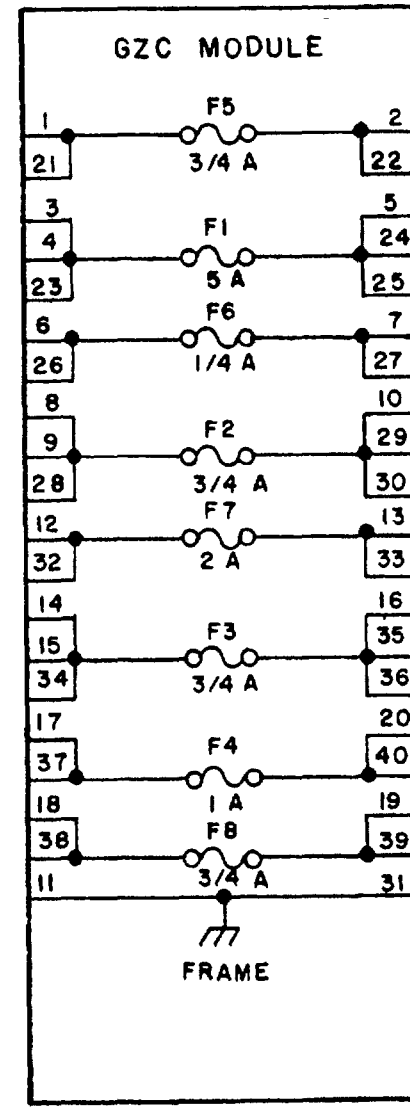
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TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	F5 Isolation	1	2	10	∞	MΩ	4
A2	F5 "	21	22	"	"	"	"
A3	F1 "	3	5	"	"	"	"
A4	F1 "	4	24	"	"	"	"
A5	F1 "	23	25	"	"	"	"
A6	F6 "	6	7	"	"	"	"
A7	F6 "	26	27	"	"	"	"
A8	F2 "	8	10	"	"	"	"
A9	F2 "	9	29	"	"	"	"
A10	F2 "	28	30	"	"	"	"
A11	F7 "	12	13	"	"	"	"
A12	F7 "	32	33	"	"	"	"
A13	F3 "	14	16	"	"	"	"
A14	F3 "	15	35	"	"	"	"
A15	F3 "	34	36	"	"	"	"
A16	F4 "	17	20	"	"	"	"
A17	F4 "	37	40	"	"	"	"
A18	F8 "	18	19	"	"	"	"
A19	F8 "	38	39	"	"	"	"

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For acceptance testing, see table I, footnote 1, and 4.4.2. For qualification testing, one end of each fuse shall be disconnected (see 4.5.1 and 4.5.2).

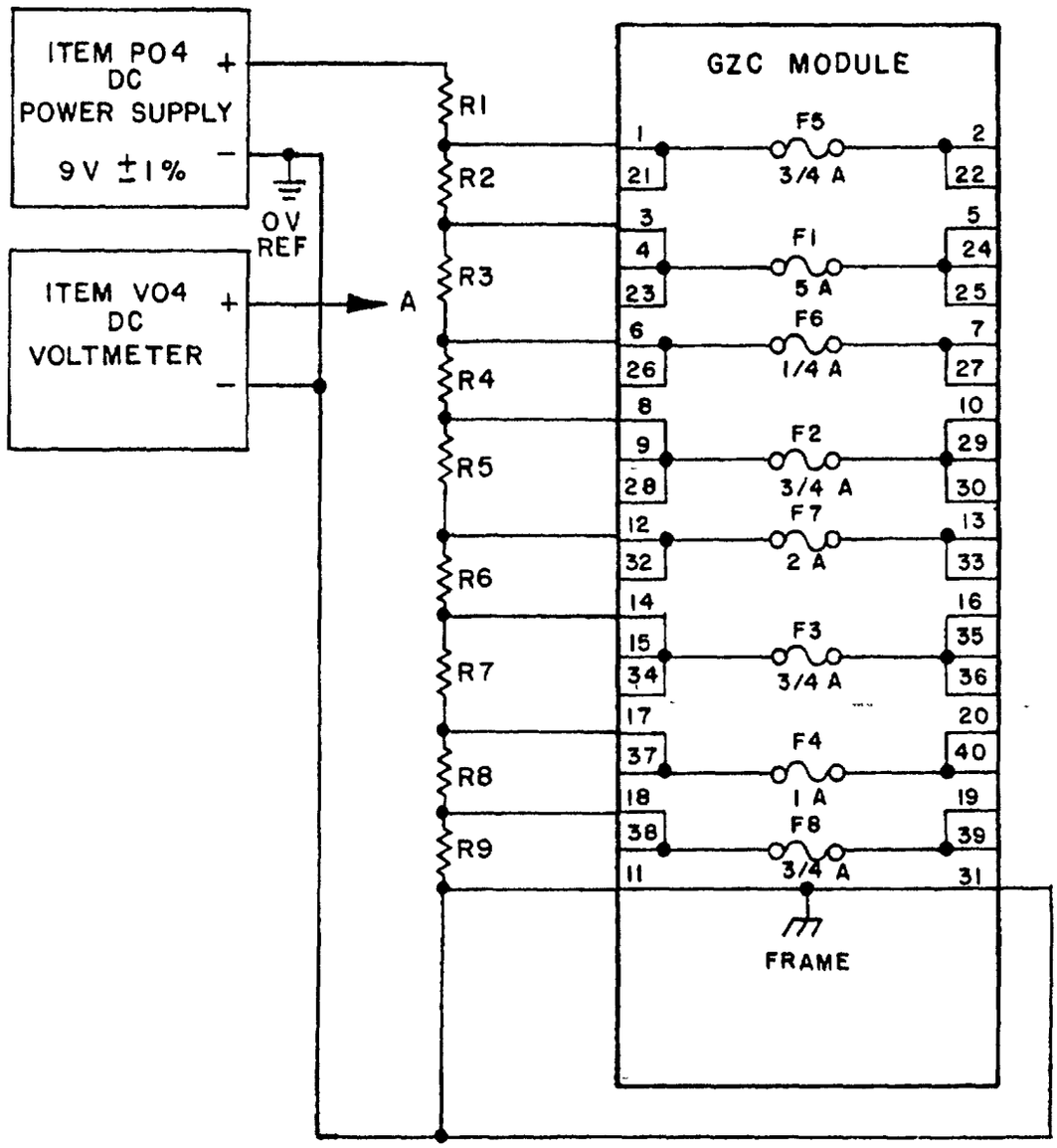
* FIGURE 4. Isolation test circuit for table II.

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TABLE III. Fuse isolation and integrity test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
C1	F5 Fuse isolation and integrity	2	7.9	8.1	V	5
C2	F1 "	5	6.9	7.1	"	"
C3	F6 "	7	5.9	6.1	"	"
C4	F2 "	10	4.9	5.1	"	"
C5	F7 "	13	3.9	4.1	"	"
C6	F3 "	16	2.9	3.1	"	"
C7	F4 "	20	1.9	2.1	"	"
C8	F8 "	19	0.9	1.1	"	"

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NOTE: R1-R9 = 1 kΩ ±1%, 1/4 W.

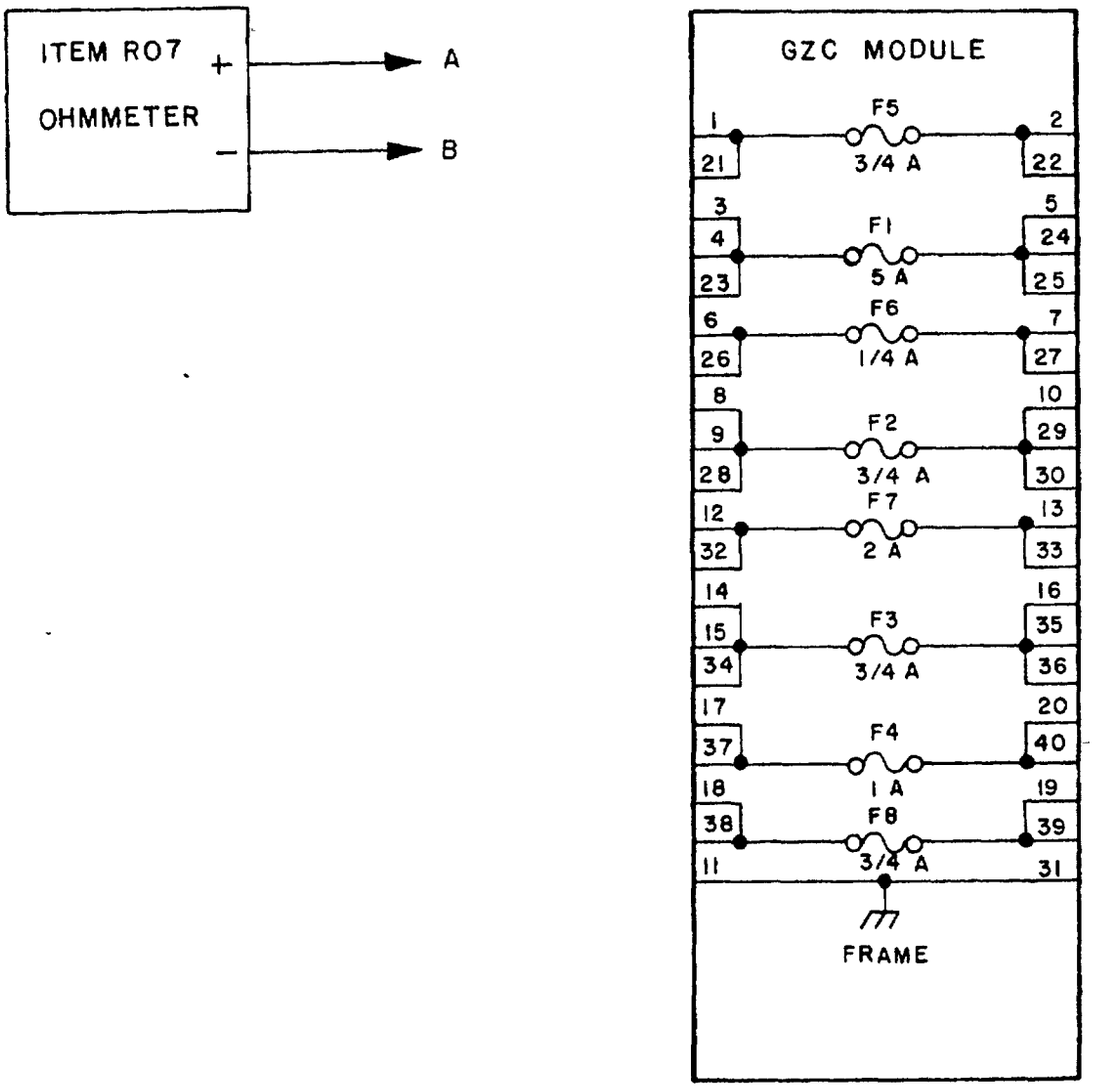
* FIGURE 5. Fuse isolation and integrity test circuit for table III.

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TABLE IV. Continuity (fuse and frame pins) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	F5 Continuity, fuse	1	2	0	1	Ω	6
D2	F5 "	21	22	0	1	"	"
D3	F6 "	6	7	0.0	1.6	"	"
D4	F6 "	26	27	0.0	1.6	"	"
D5	F7 "	12	13	0	1	"	"
D6	F7 "	32	33	"	"	"	"
D7	F1 "	3	24	"	"	"	"
D8	F1 "	4	5	"	"	"	"
D9	F1 "	23	25	"	"	"	"
D10	F2 "	8	10	"	"	"	"
D11	F2 "	9	29	"	"	"	"
D12	F2 "	28	30	"	"	"	"
D13	F3 "	14	16	"	"	"	"
D14	F3 "	15	35	"	"	"	"
D15	F3 "	34	36	"	"	"	"
D16	F4 "	17	20	"	"	"	"
D17	F4 "	37	40	"	"	"	"
D18	F8 "	18	19	"	"	"	"
D19	F8 "	38	39	"	"	"	"
D20	Continuity, frame pins	11	31	"	"	"	"

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* FIGURE 6. Continuity (fuse and frame pins) test circuit for table IV.

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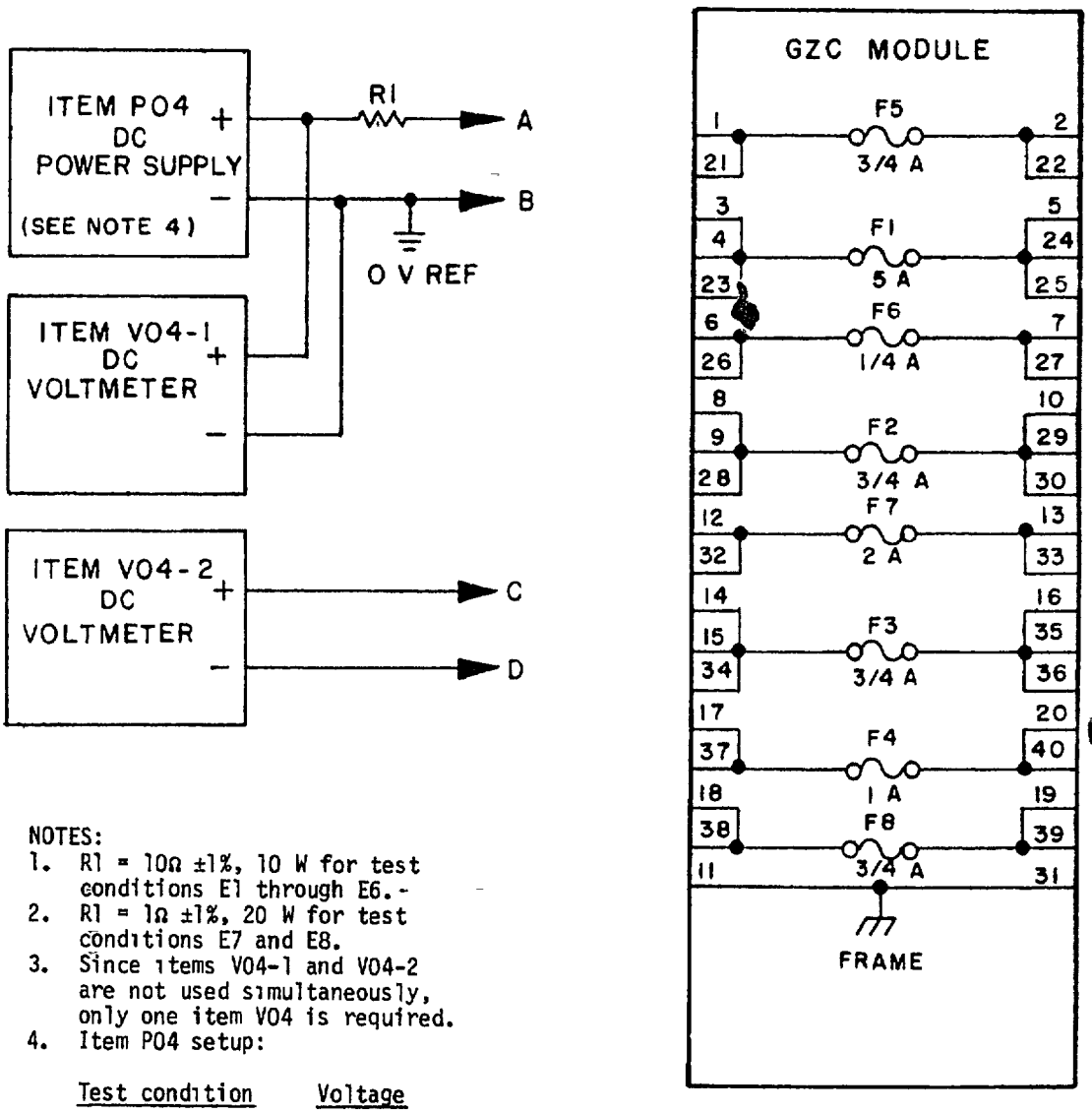
TABLE V. Voltage drop test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
E1	F5 Voltage drop	21	22	1	2	70	180	mV	7
E2	F2 "	9	29	8	10	"	"	"	"
E3	F3 "	15	35	14	16	"	"	"	"
E4	F8 "	38	39	18	19	"	"	"	"
E5	F6 "	26	27	6	7	"	375	"	"
E6	F4 "	37	40	17	20	"	180	"	"
E7	F7 "	32	33	12	13	"	"	"	"
E8	F1 "	4	25	3	5	"	"	"	"

TABLE VI. Continuity test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
F1	Continuity	11	Frame	0	1	Ω	8
F2	Continuity	31	Frame	0	1	Ω	8

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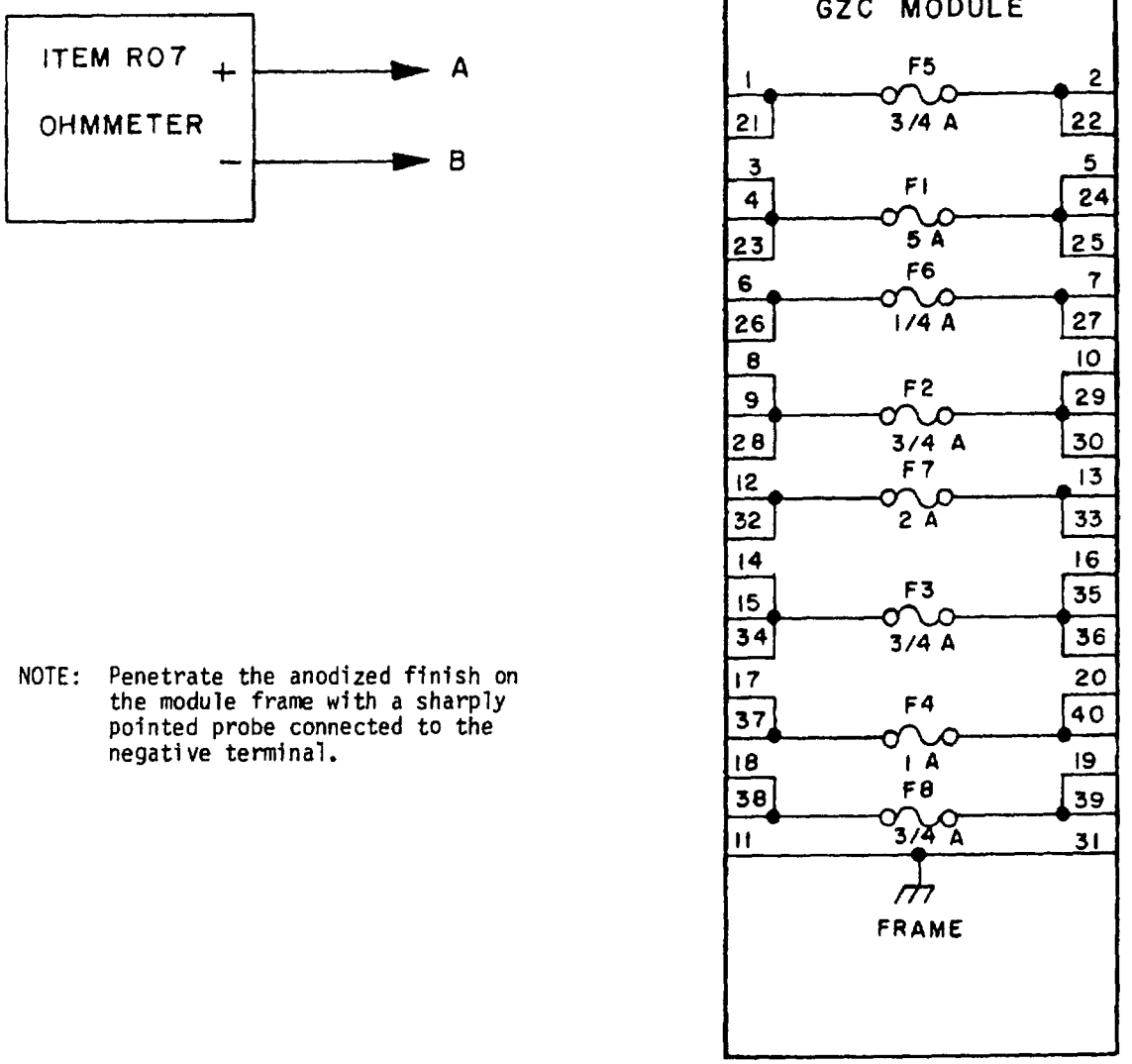


NOTES:
1. R1 = 10Ω ±1%, 10 W for test conditions E1 through E6.
2. R1 = 1Ω ±1%, 20 W for test conditions E7 and E8.
3. Since items V04-1 and V04-2 are not used simultaneously, only one item V04 is required.
4. Item P04 setup:

Test condition	Voltage
E1-E4	5.60 ±0.06 V
E5	1.80 ±0.02 V
E6	7.50 ±0.08 V
E7	1.50 ±0.02 V
E8	3.70 ±0.04 V

* FIGURE 7. Voltage drop test circuit for table V.

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NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

* FIGURE 8. Continuity test circuit for table VI.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28787/209B(EC)	Modules, Standard Electronic Fuse, Key Code GZC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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